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Experiment 4: Provision a VPS instance from a cloud provider or set up a virtual machine on your local system using software like VirtualBox or VMware. 2. Install a Linux distribution such as Ubuntu, CentOS, or Debian as the operating system. Configure network settings, including IP address, subnet mask, gateway, and DNS.

AIM:

To provision a Virtual Private Server (VPS) using VirtualBox, install Ubuntu Server, configure network settings (IP Address, Subnet Mask, Gateway, DNS), and enable SSH remote access.

What is VPS?

A Virtual Private Server (VPS) is a software-based server created using virtualization technology. It operates independently with its own operating system, storage, memory, and network configuration while sharing the physical hardware with other virtual machines. Virtualization allows efficient utilization of resources and forms the foundation of cloud computing platforms.

Network configuration parameters include:

IP Address

A unique address assigned to a device on a network.

Subnet Mask

Determines the network and host portions of an IP address.

Gateway

Provides access to networks outside the local network.

DNS (Domain Name System)

Converts domain names into IP addresses.

SSH (Secure Shell)

A secure protocol used for remote administration of Linux systems.

PROCEDURE:

1. Open the Ubuntu terminal and verify the internet connection using the ping google.com command.

```

sudha@ubuntu-server:~$ ping google.com
PING google.com (2404:6800:4007:808::200e) 56 data bytes
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=1 ttl=117 time=5.13 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=2 ttl=117 time=21.3 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=3 ttl=117 time=15.6 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=4 ttl=117 time=16.6 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=5 ttl=117 time=16.2 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=6 ttl=117 time=8.58 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=7 ttl=117 time=15.8 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=8 ttl=117 time=15.1 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=9 ttl=117 time=20.0 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=10 ttl=117 time=16.2 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=11 ttl=117 time=12.0 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=12 ttl=117 time=16.6 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=13 ttl=117 time=17.3 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=14 ttl=117 time=20.5 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=15 ttl=117 time=19.6 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=16 ttl=117 time=16.3 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=17 ttl=117 time=16.6 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=18 ttl=117 time=19.5 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=19 ttl=117 time=17.4 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=20 ttl=117 time=12.5 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=21 ttl=117 time=15.7 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=22 ttl=117 time=8.54 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=23 ttl=117 time=19.1 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=24 ttl=117 time=7.43 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=25 ttl=117 time=25.1 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=26 ttl=117 time=6.40 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=27 ttl=117 time=19.6 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=28 ttl=117 time=19.5 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=29 ttl=117 time=15.7 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=30 ttl=117 time=16.5 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=31 ttl=117 time=5.48 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=32 ttl=117 time=16.7 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=33 ttl=117 time=15.7 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=34 ttl=117 time=25.6 ms
^[[S64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=35 ttl=117 time=18.0 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=36 ttl=117 time=16.0 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=37 ttl=117 time=19.5 ms
^[[S64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=38 ttl=117 time=14.6 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=39 ttl=117 time=14.5 ms
64 bytes from pnmaaa-aq-in-x0e.1e100.net (2404:6800:4007:808::200e): icmp_seq=40 ttl=117 time=16.3 ms
^C

```

2. Check the network routing information using the ip route command.

```

rtt min/avg/max/mdev = 5.132/15.866/25.642/4.636 ms
sudha@ubuntu-server:~$ ip route
default via 192.168.0.1 dev enp0s3 proto dhcp src 192.168.0.5 metric 100
192.236.180.15 via 192.168.0.1 dev enp0s3 proto dhcp src 192.168.0.5 metric 100
192.168.0.0/24 dev enp0s3 proto kernel scope link src 192.168.0.5 metric 100
192.168.0.1 dev enp0s3 proto dhcp scope link src 192.168.0.5 metric 100
sudha@ubuntu-server:~$

```

3. Verify the DNS (Domain Name System) configuration of the system using the `cat /etc/resolv.conf` command.

```

192.168.0.1 dev enp0s3 proto udp scope link src 192.168.0.1 metric 100
sudha@ubuntu-server:~$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
search .
sudha@ubuntu-server:~$

```

4. Update the Ubuntu package list (if required) using: `sudo apt update`
5. Install the OpenSSH Server package using: `sudo apt install openssh-server -y`
6. Enable and start the SSH service using:
7. `sudo systemctl enable ssh` `sudo systemctl start ssh`
8. Check whether the SSH service is running successfully using:

`sudo systemctl status`

```

sudha@ubuntu-server:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-08-29 02:16:35 UTC; 4min 59s ago
   Invocation: 9700c15cddd84fd4a5b1bda1f0e35d4e
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
         man:sshd_config(5)
   Process: 2334 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 2337 (sshd)
   Tasks: 1 (limit: 1885)
   Memory: 1.2M (peak: 1.9M)
   CPU: 69ms
   CGroup: /system.slice/ssh.service
           └─2337 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 29 02:16:35 ubuntu-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 29 02:16:35 ubuntu-server sshd[2337]: Server listening on 0.0.0.0 port 22.
Aug 29 02:16:35 ubuntu-server sshd[2337]: Server listening on :: port 22.
Aug 29 02:16:35 ubuntu-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
sudha@ubuntu-server:~$

```

9. Find the IP address of the Ubuntu system using:

Hostname -I

10. Establish an SSH connection to the Ubuntu system using:

ssh <username>@<IP_address>

```
sudha@ubuntu-server:~$ hostname -I
192.168.0.5 2406:7400:c4:d056:a00:27ff:fe17:b629
sudha@ubuntu-server:~$ ssh sudha@192.168.0.5
The authenticity of host '192.168.0.5 (192.168.0.5)' can't be established
ED25519 key fingerprint is: SHA256:s1I1E5XjZ7dRc4yrPtwwSwvkS3NpFThpKimIlW
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.5' (ED25519) to the list of known h
sudha@192.168.0.5's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Aug 29 02:25:22 AM UTC 2026

System load:            0.0
Usage of /:              44.2% of 11.21GB
Memory usage:           17%
Swap usage:             0%
Processes:              106
Users logged in:        0
IPv4 address for enp0s3: 192.168.0.5
IPv6 address for enp0s3: 2406:7400:c4:d056:a00:27ff:fe17:b629

Expanded Security Maintenance for Applications is not enabled.

2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

11. Verify the successful SSH login by checking the "Welcome to Ubuntu" message displayed on the terminal.

CONCLUSION:

The Ubuntu system was successfully configured and verified for network connectivity. The OpenSSH Server was installed and enabled, and the SSH service was confirmed to be running. The system's IP address was obtained, and a successful SSH login was performed. Thus, secure remote access to the Ubuntu system was successfully established and verified.